Cold-pool propagation: from point arrivals to a source

Input: 197 original 5-minute cold-pool candidates, including 9 Eq. (3) episodes.
The newer 1-minute and modified-monotonicity catalogs are not used.

Three models: expanding circle; advected circle; advected ellipse.

Goal: infer release location and time from station arrival times.
First validate the inverse method using synthetic data with a known source.

Real-data fits are conditional hypotheses, not confirmed origins.
Association, propagation speed, wind translation and terrain remain uncertain.

Forward equations and assumptions

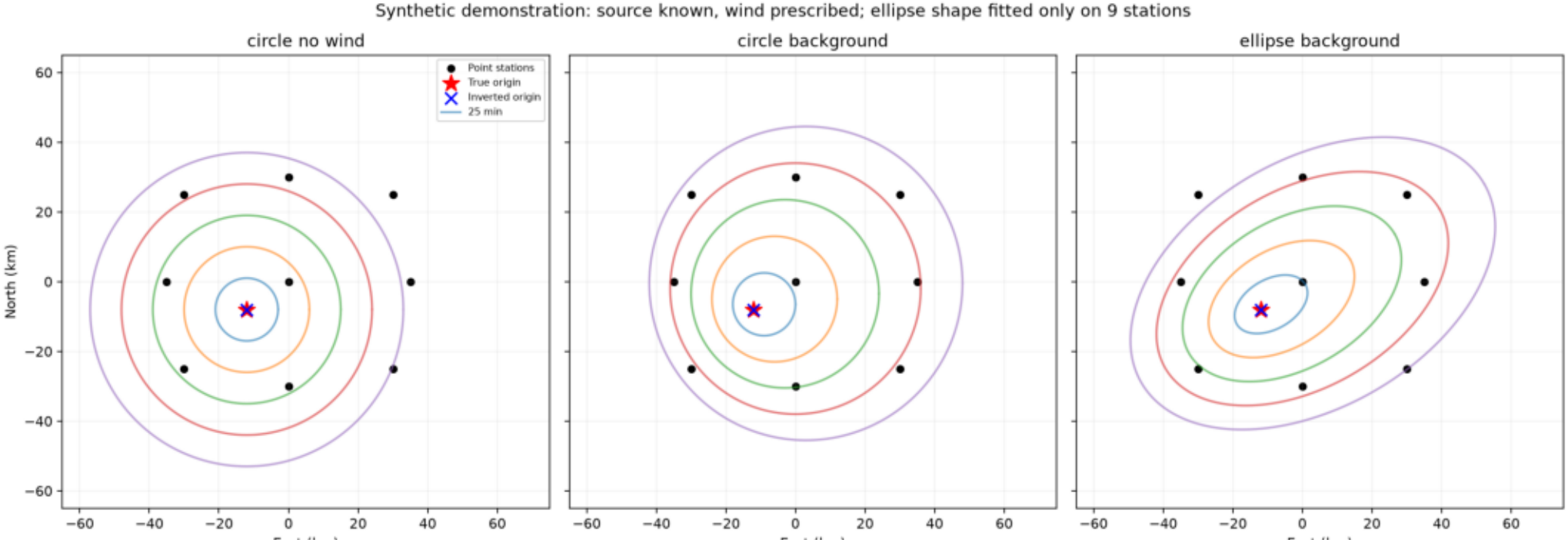
Circle, no wind: $|r_i - r_0| = c\,(t_i - t_0)$.
Unknown source $r_0=(x_0,y_0)$, release time t_0, and radial speed c.

With wind: $\text{center}(t)=r_0+U(t-t_0)$; $\text{radius}=c(t-t_0)$.
U is prescribed translation; measured surface wind is only a proxy.

Ellipse: semi-axes grow at $c\sqrt{r}$ and $c/\sqrt{r}$, with fixed orientation theta.
r is the major/minor axis ratio. $r=1$ exactly recovers the circle.

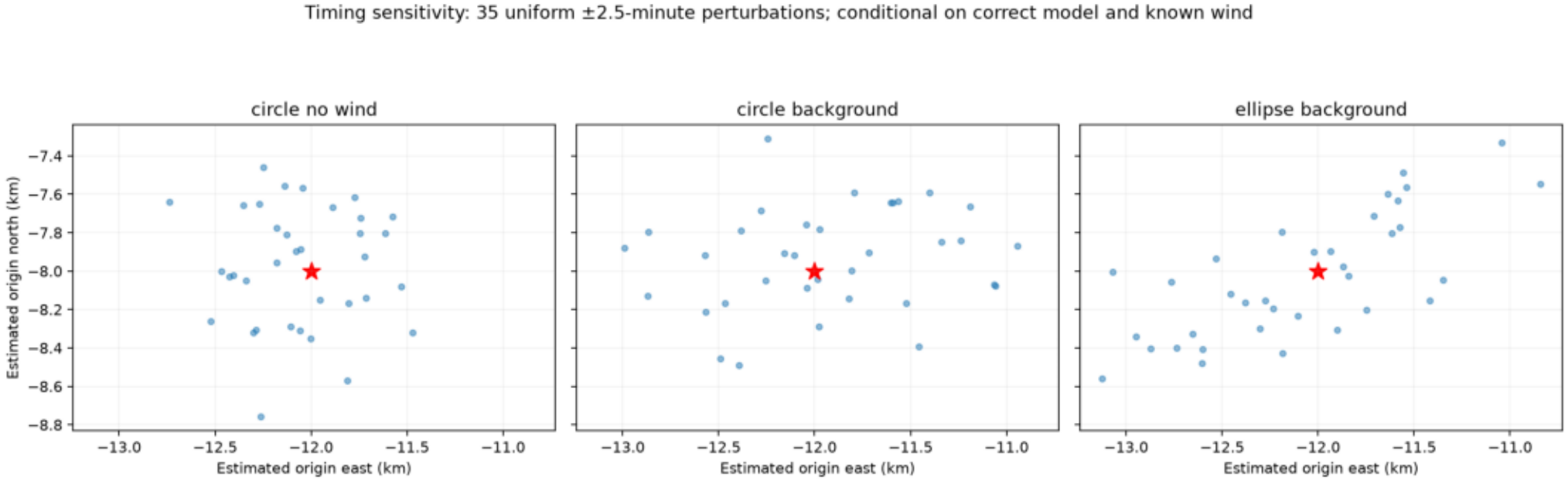
Constant speeds and wind; flat horizontal geometry; no terrain dynamics.
Kinematic conceptual models, not full atmospheric fluid simulations.

Three conceptual stages and known-source recovery



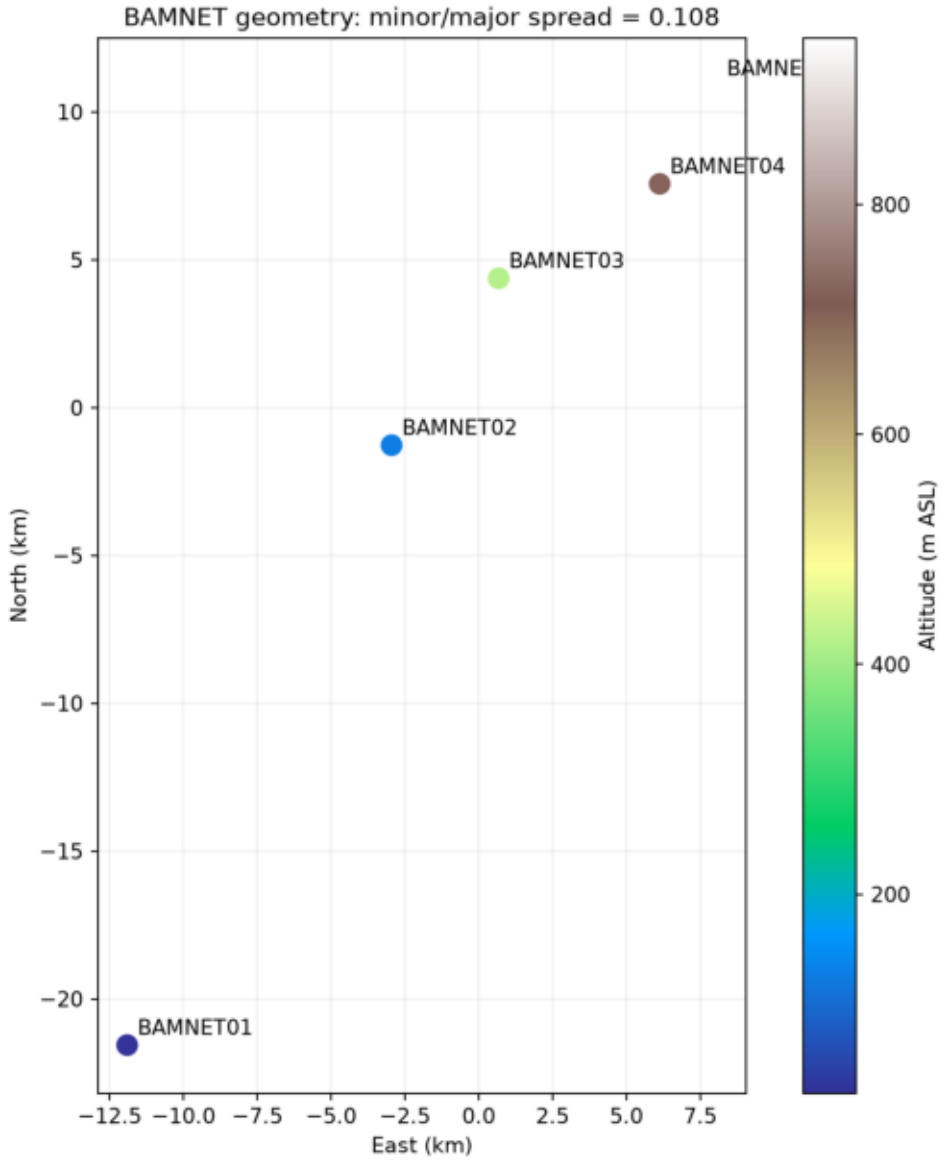
Nine distributed synthetic stations; prescribed wind; free ellipse fit only in this synthetic demonstration.

Sensitivity to five-minute timing resolution



35 independent uniform ±2.5-minute perturbations per model. Conditional sensitivity, not real-data confidence intervals.

Actual station geometry and elevation



Minor/major spatial spread ratio 0.108; elevation 27–945 m ASL. Horizontal model ignores terrain.

What can be identified from point observations?

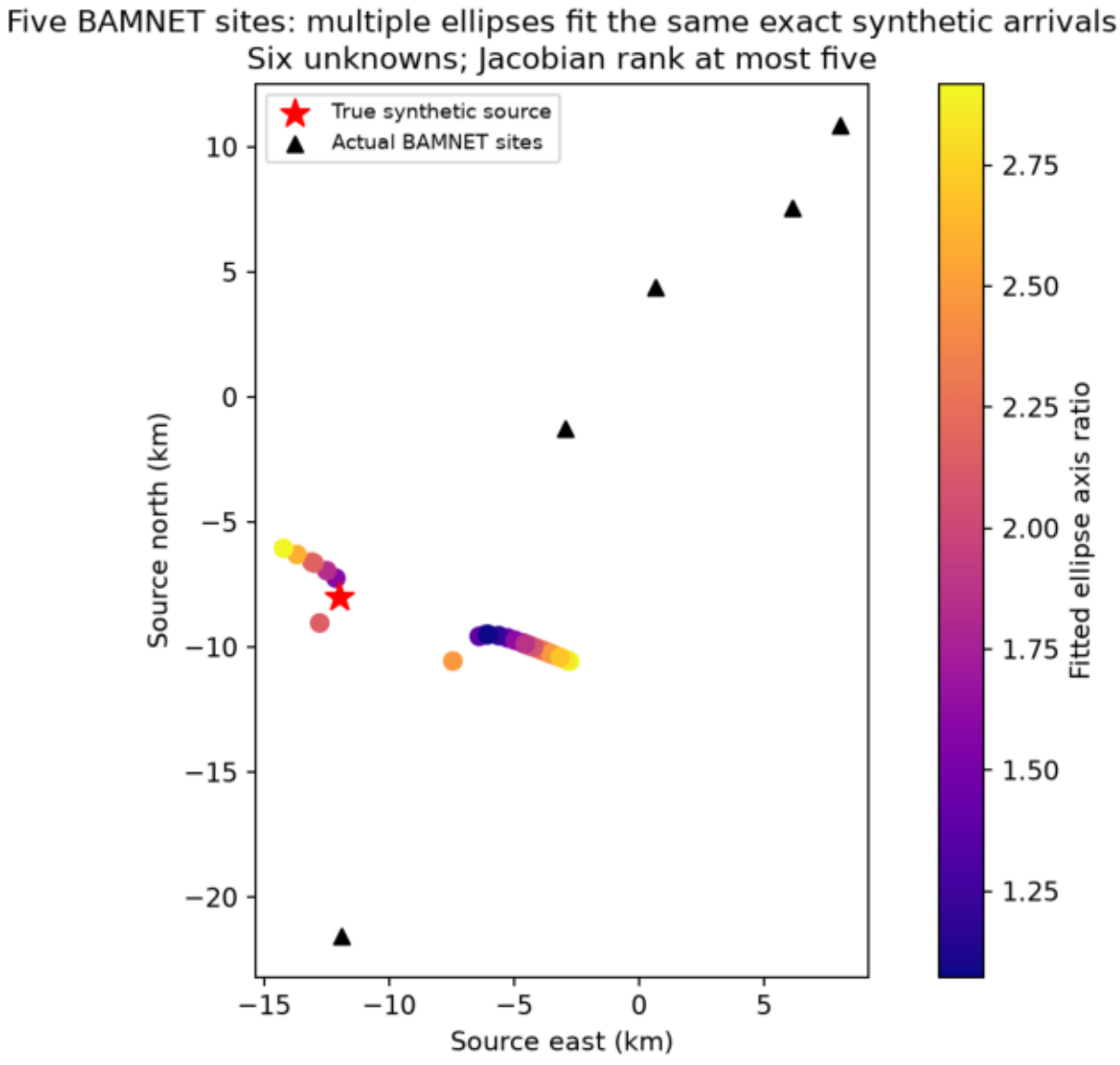
Circle with unknown speed: 4 unknowns (x,y,t0,c).
Four arrivals give zero residual degrees of freedom, not guaranteed uniqueness.

Free ellipse with prescribed wind: 6 unknowns.
Five single-pass station arrivals cannot identify all six parameters.

Real-data main fits fix c=5 m/s and shape assumptions: 3 source unknowns.
Sensitivity uses c=3 and 8 m/s; speed is a scenario, not an observed quantity.

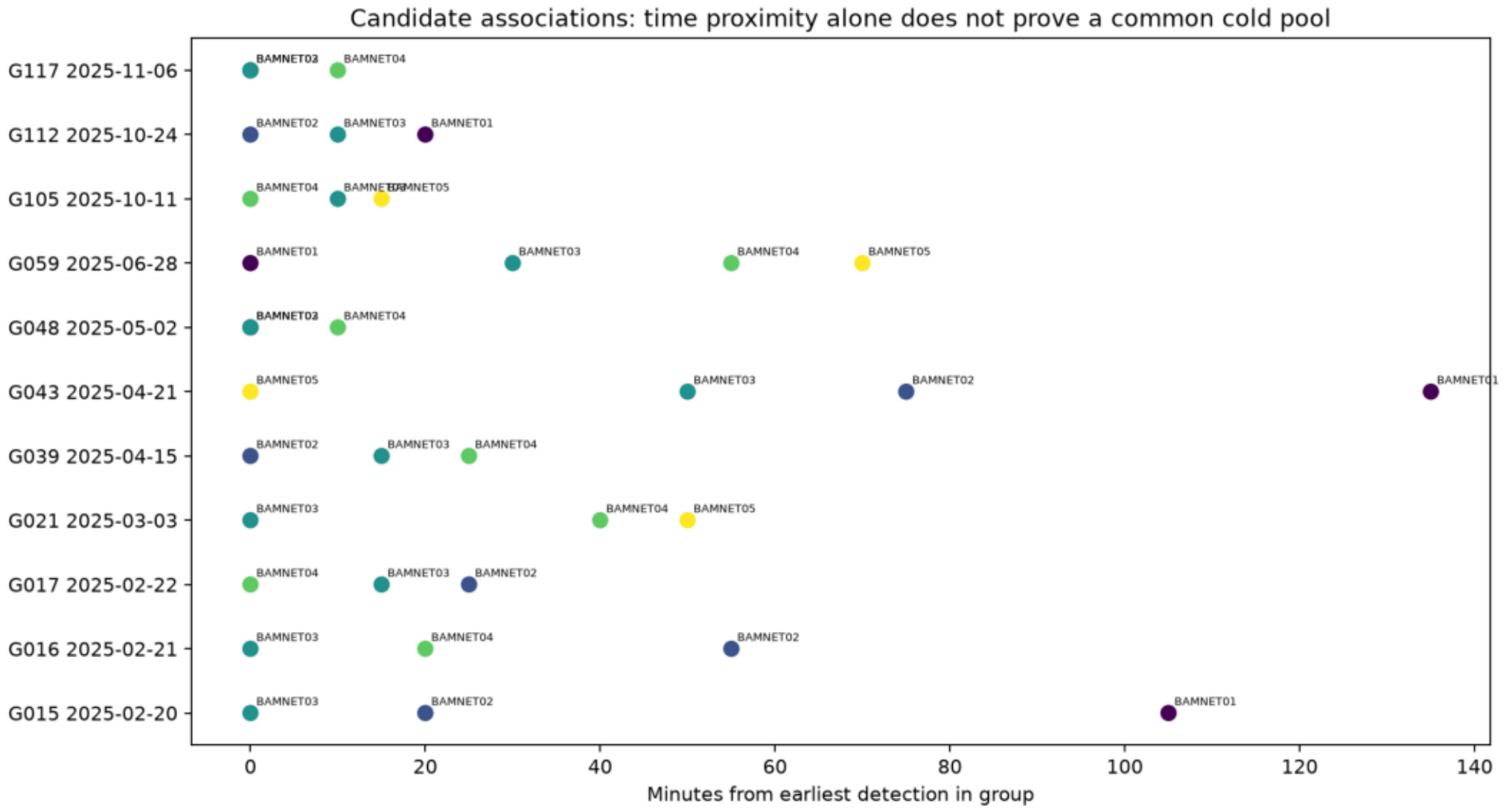
Unknown background wind adds two more unknowns.
Three observations with three source parameters also leave no residual df.

Same exact arrivals, different free ellipses



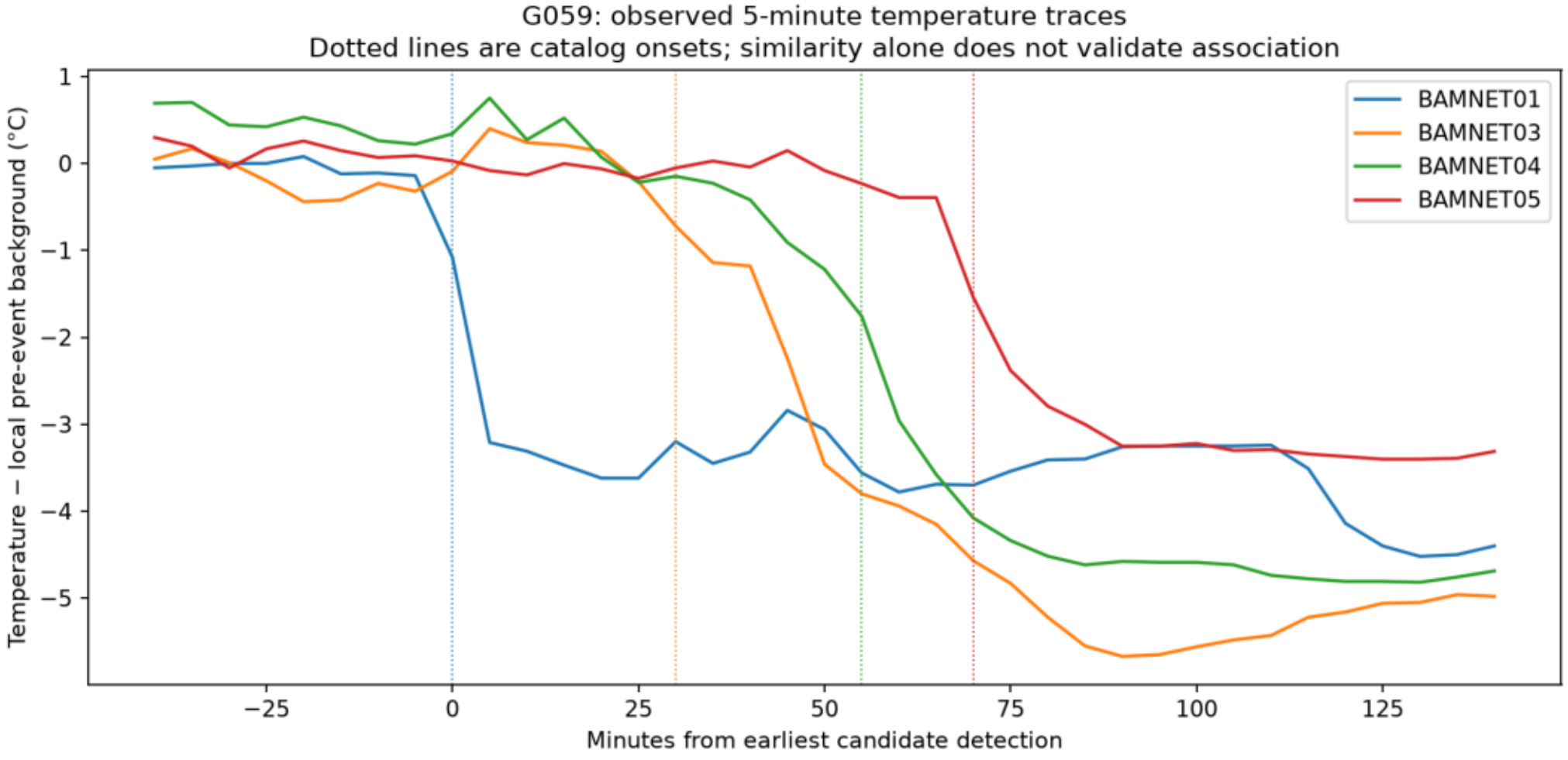
35 near-exact solutions on the five actual site coordinates. This is synthetic, not a real recovered source.

Temporal association is a hypothesis



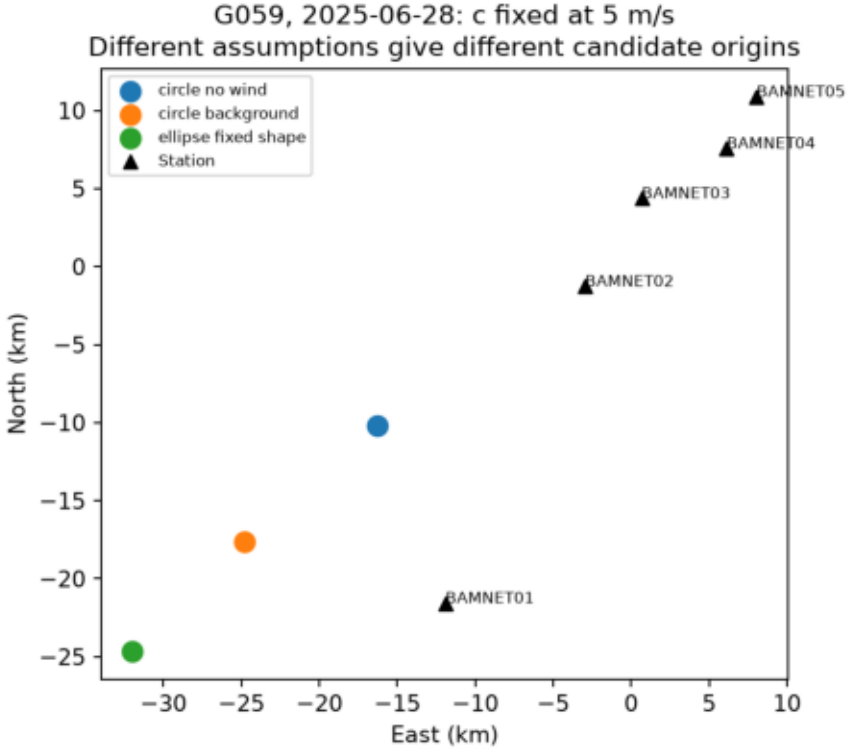
Primary bounded window: 180 min from first detection. Eleven groups have ≥3 sites; two have four; none has five.

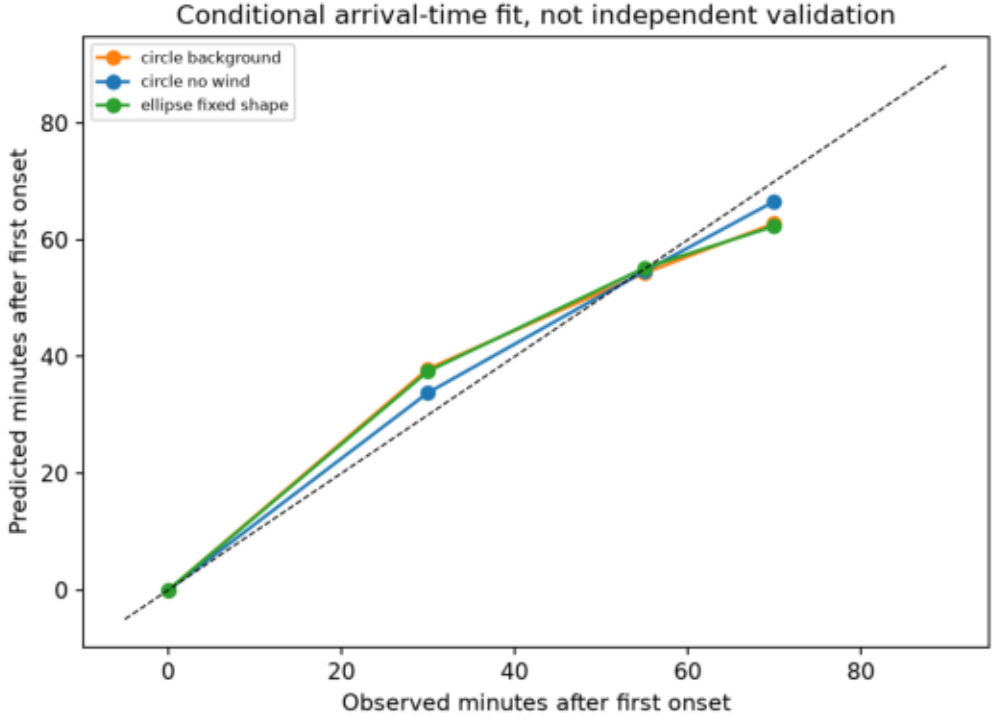
Inspect the actual observations



Background-normalized temperature helps inspect association, but common timing and cooling do not prove a single source.

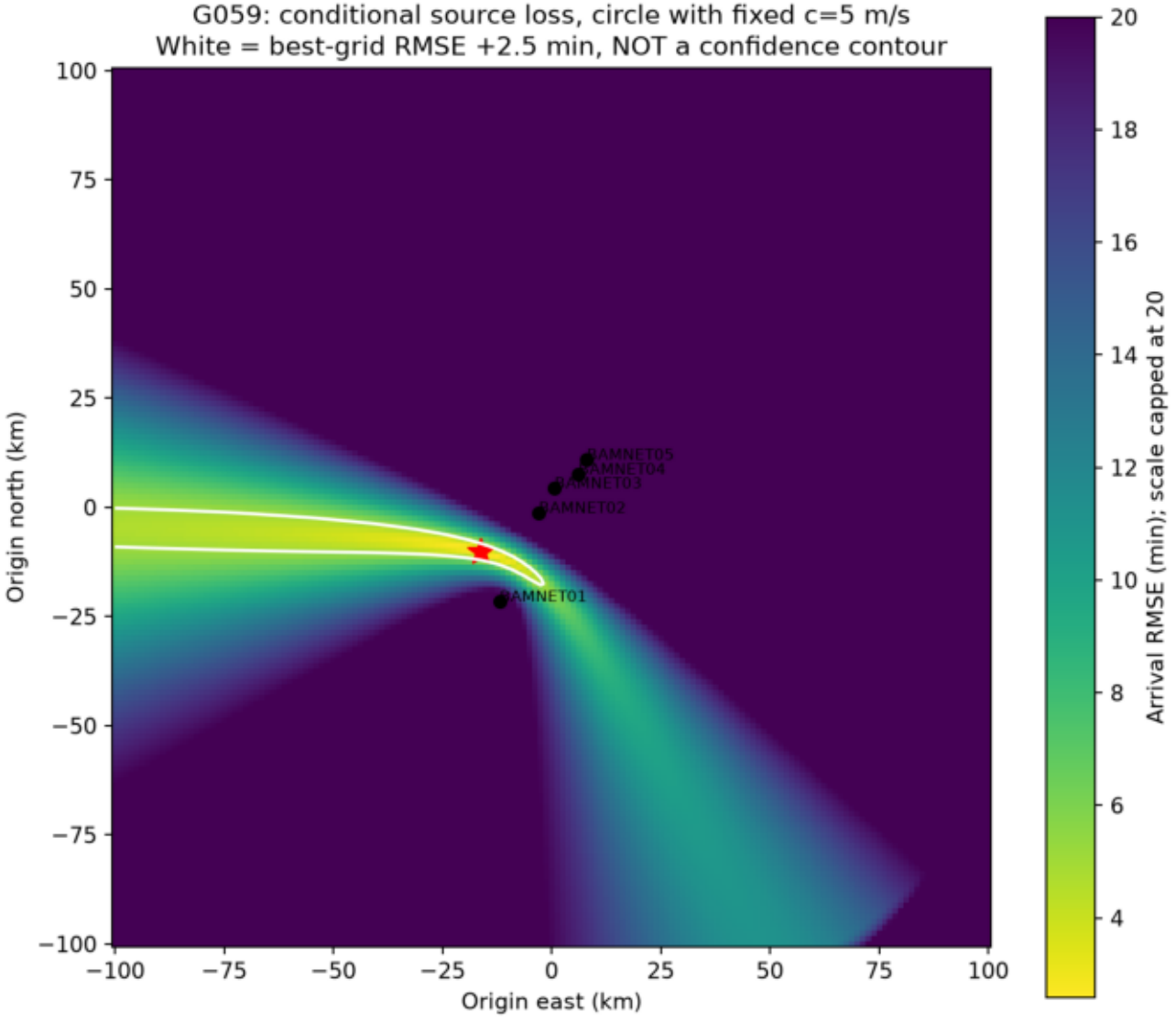
Conditional real-data example: 28 June 2025





Fixed c=5 m/s. Prescribed background wind/shape alter the inferred origin. No model is declared the true source.

Source-loss surface, not a confidence map



Circle with c=5 m/s. White contour is best grid RMSE +2.5 min; not a 95% confidence boundary.

What to present at the next meeting

Synthetic recovery works under its stated assumptions; real-data origins remain conditional.

The 28 June group has a circle/no-wind c=5 m/s RMSE of about 2.56 min.
Fitting speed as well can drive RMSE to zero with four data and four unknowns.

Zero residual is not validation. Another four-station case hits prior bounds
and has larger residuals: constant speed, association, or model may be inappropriate.

Next: independently verify shared events, constrain propagation speed/wind,
add off-axis observations, and use radar/satellite/terrain information.

GIFs, all fit tables, alternative solutions, source-loss arrays, inputs and code
are included. Earlier detection files have not been changed.